Simple Search Engine using Hadoop MapReduce, Cassandra, and Spark

Assignment 2 Report

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Repository	https://github.com/Ily17as/big-data-assignment2-submission.git
Environment	WSL2 + Docker Compose + Hadoop/Spark + Cassandra
Indexed corpus size	1000 documents from a.parquet

1. Methodology

The project implements a simple batch-oriented search engine for plain-text documents. The pipeline is intentionally modular: PySpark prepares the document corpus, Hadoop Streaming builds the index in HDFS, Cassandra stores the index for retrieval, and Spark on YARN executes BM25 ranking for user queries.

The final submission repository used for this report is big-data-assignment2-submission. The grading model described in the assignment is satisfied by a single docker compose up –-build run from a clean clone, and the same run also demonstrates the optional add_to_index extension.

1.1 End-to-End Workflow

1. start-services.sh starts Hadoop services and prepares the distributed environment;
2. prepare_data.py reads a.parquet, filters valid rows, builds 1000 UTF-8 text documents, and writes /input/data as a single-partition HDFS dataset;
3. prepare_data.sh uploads the plain-text document corpus to HDFS /data and prints explicit upload progress to make long-running uploads observable;
4. create_index.sh runs Hadoop Streaming with mapper1.py and reducer1.py, produces raw tagged records, and splits them into the HDFS folders /indexer/documents, /indexer/postings, /indexer/stats, and /indexer/vocabulary;

5. store_index.sh exports these HDFS partitions and loads them into Cassandra tables;
6. search.sh submits query.py to YARN, where BM25 is computed and the top 10 results are printed.
7. add_to_index.sh accepts a local text file, inserts it into the indexed collection, rebuilds the input and index, reloads Cassandra, and validates the new document by search.

Implementation Snapshot

Component	Implementation	Notes
Data preparation	PySpark	Reads a.parquet, selects id/title/text, writes /input/data and UTF-8 text files under /data.
Indexer	Hadoop Streaming	One MapReduce pipeline emits DOC, VOCAB, POSTING, and STAT records; create_index.sh splits them into /indexer subfolders.
Storage	Cassandra keyspace search_engine	Tables: documents, vocabulary, postings, corpus_stats.
Ranker	PySpark on YARN	query.py computes BM25 with k1=1.5 and b=0.75 and returns top 10 results.
Extra task	add_to_index.sh	Adds a local file, rebuilds /input/data, recreates the index, reloads Cassandra, and validates retrieval.

1.2 Data Preparation

- Input source: a.parquet in HDFS.
- Selected fields: id, title, text.
- Prepared corpus size for the main run: 1000 documents.
- Text normalization: plain UTF-8 text with document identifiers and titles carried into the retrieval layer.
- HDFS outputs: /data for document files and /input/data for the tab-separated indexer input.

A practical change in the final version was the introduction of explicit upload progress logging for the /data stage. This makes the pipeline easier to monitor and also provides a useful demonstration artifact for the report.

1.3 Indexer Design

- The Hadoop Streaming mapper/reducer emit tagged lines with four record types: DOC, VOCAB, POSTING, and STAT.
- create_index.sh first writes the raw reducer output to /tmp/indexer/raw and then materializes the final HDFS index into separate folders under /indexer.
- This design keeps the indexing logic simple while preserving a clear physical separation between document metadata, term document frequencies, postings, and corpus-level statistics needed for BM25.

The resulting HDFS layout is easy to inspect and export: /indexer/documents contains document titles and lengths, /indexer/postings contains term-document tf entries, /indexer/vocabulary contains document frequency per term, and /indexer/stats contains global values such as N and AVGDL.

1.4 Cassandra Storage Schema

Table	**Primary key**	**Stored fields**
documents	doc_id	title, doc_len
vocabulary	term	df
postings	(term, doc_id)	tf
corpus_stats	stat_key	stat_value

The Cassandra schema is intentionally minimal and directly matches what query.py needs at search time. This avoids expensive joins and keeps the online retrieval logic straightforward.

1.5 Retrieval Model

- Tokenization uses the regular expression [A-Za-z0-9]+ and lowercasing.
- BM25 parameters in query.py are k1 = 1.5 and b = 0.75.
- The ranker fetches N and AVGDL from corpus_stats, df values from vocabulary, and tf postings from postings, then computes per-term BM25 contributions and aggregates them with Spark RDD operations.
- The final output is the top 10 documents sorted by descending BM25 score.

Observed Results from the Successful Run

Metric	Observed value
Prepared documents	1000 documents created from a.parquet and transformed into /input/data for indexing.
Upload progress	The /data upload stage reports progress from 1/1000 to 1000/1000 with zero failed files. If you upload the documents from the repository operation can fail due to file name is not in latin namespace (there is non-latin characters in the name). I tried to normalize the names before uploading it, but, unfortunately, normalization cannot cover all cases.
Index partitions in HDFS	documents, postings, stats, and vocabulary were written as single partitions under /indexer.
Final index sizes after extra task	documents: 43,405 B; postings: 4,371,358 B; stats: 31 B; vocabulary: 2,198,514 B.
Search queries demonstrated	computer science, artificial intelligence, and galievtesttoken.
Extra task outcome	The newly added document 99999999_GalievTestToken.txt became the top result for query galievtesttoken.

2. Demonstration

The repository is designed so that the full baseline can be reproduced with a clean environment using docker compose up --build from the project root. During the successful runs used for this report, the pipeline completed end-to-end: 1000 documents were prepared, uploaded to HDFS, indexed in HDFS, loaded into Cassandra, and searched through YARN-based Spark jobs.

The following terminal captures are included as fullscreen evidence of the final working system. The first screenshot confirms successful indexing, the next two show successful query execution, and the last screenshot confirms the optional add_to_index task.

2.1 How to Run

```
docker compose down -v
```

```
docker compose up --build
```

The final app.sh run demonstrates the baseline queries "computer science" and "artificial intelligence". The extra-task validation adds a new document with a unique token and then searches for that token.

2.2 Evidence from the Successful Runs

```
ilyasgaliev@MSI: ~/bigdata/bi
afddfb9dbc1e30688056eed6c302ba09/nb-6-big
cassandra-server  | INFO  [Native-Transport-Requests-8] 2026-03-29 15:18:12,710 QueryProcessor.java:700 - Fully upgraded to at least 5.0.6
cluster-master    | Index loaded into Cassandra successfully.
cluster-master    | Indexing pipeline completed successfully.
cluster-master    | Running BM25 search on YARN
cluster-master    | 26/03/29 15:24:39 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where appl
icable
cluster-master    | 26/03/29 15:24:39 INFO SparkContext: Running Spark version 3.5.4
cluster-master    | 26/03/29 15:24:39 INFO SparkContext: OS info Linux, 6.6.87.2-microsoft-standard-WSL2, amd64
cluster-master    | 26/03/29 15:24:39 INFO SparkContext: Java version 1.8.0_442
cluster-master    | 26/03/29 15:24:40 INFO ResourceUtils: ==============================================================
cluster-master    | 26/03/29 15:24:40 INFO ResourceUtils: No custom resources configured for spark.driver.
cluster-master    | 26/03/29 15:24:40 INFO ResourceUtils: ==============================================================
cluster-master    | 26/03/29 15:24:40 INFO SparkContext: Submitted application: bm25-query
cluster-master    | 26/03/29 15:24:40 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name: cores, amount: 1, script
: , vendor: , memory -> name: memory, amount: 1024, script: , vendor: , offHeap -> name: offHeap, amount: 0, script: , vendor: ), task resources: Map(cpus -
> name: cpus, amount: 1.0)
cluster-master    | 26/03/29 15:24:40 INFO ResourceProfile: Limiting resource is cpus at 1 tasks per executor
cluster-master    | 26/03/29 15:24:40 INFO ResourceProfileManager: Added ResourceProfile id: 0
cluster-master    | 26/03/29 15:24:40 INFO SecurityManager: Changing view acls to: root
cluster-master    | 26/03/29 15:24:40 INFO SecurityManager: Changing modify acls to: root
cluster-master    | 26/03/29 15:24:40 INFO SecurityManager: Changing view acls groups to:
cluster-master    | 26/03/29 15:24:40 INFO SecurityManager: Changing modify acls groups to:
cluster-master    | 26/03/29 15:24:40 INFO SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users with view permissions: root; g
roups with view permissions: EMPTY; users with modify permissions: root; groups with modify permissions: EMPTY
cluster-master    | 26/03/29 15:24:40 INFO Utils: Successfully started service 'sparkDriver' on port 36203.
cluster-master    | 26/03/29 15:24:40 INFO SparkEnv: Registering MapOutputTracker
cluster-master    | 26/03/29 15:24:40 INFO SparkEnv: Registering BlockManagerMaster
cluster-master    | 26/03/29 15:24:40 INFO BlockManagerMasterEndpoint: Using org.apache.spark.storage.DefaultTopologyMapper for getting topology information
cluster-master    | 26/03/29 15:24:40 INFO BlockManagerMasterEndpoint: BlockManagerMasterEndpoint up
cluster-master    | 26/03/29 15:24:40 INFO SparkEnv: Registering BlockManagerMasterHeartbeat
cluster-master    | 26/03/29 15:24:40 INFO DiskBlockManager: Created local directory at /tmp/blockmgr-c586f588-0083-4843-ae54-8bc68b09a7e1
cluster-master    | 26/03/29 15:24:40 INFO MemoryStore: MemoryStore started with capacity 366.3 MiB
cluster-master    | 26/03/29 15:24:40 INFO SparkEnv: Registering OutputCommitCoordinator
cluster-master    | 26/03/29 15:24:40 INFO JettyUtils: Start Jetty 0.0.0.0:4040 for SparkUI
cluster-master    | 26/03/29 15:24:41 INFO Utils: Successfully started service 'SparkUI' on port 4040.
cluster-master    | 26/03/29 15:24:41 INFO DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at cluster-master/172.18.0.4:8032
cluster-master    | 26/03/29 15:24:41 INFO Configuration: resource-types.xml not found
cluster-master    | 26/03/29 15:24:41 INFO ResourceUtils: Unable to find 'resource-types.xml'.
cluster-master    | 26/03/29 15:24:41 INFO Client: Verifying our application has not requested more than the maximum memory capability of the cluster (8192
MB per container)
cluster-master    | 26/03/29 15:24:41 INFO Client: Will allocate AM container, with 896 MB memory including 384 MB overhead
cluster-master    | 26/03/29 15:24:41 INFO Client: Setting up container launch context for our AM
cluster-master    | 26/03/29 15:24:41 INFO Client: Setting up the launch environment for our AM container
```

Figure 1. The indexing pipeline completed successfully and the ranker immediately started BM25 search on YARN.

```
ilyasgaliev@MSI: ~/bigdata/bi
.0.2:42550) with ID 1,  ResourceProfileId 0
cluster-master    | 26/03/29 15:25:09 INFO YarnSchedulerBackend$YarnDriverEndpoint: Registered executor NettyRpcEndpointRef(spark-client://Executor) (172.18
.0.2:42558) with ID 2,  ResourceProfileId 0
cluster-master    | 26/03/29 15:25:09 INFO YarnClientSchedulerBackend: SchedulerBackend is ready for scheduling beginning after reached minRegisteredResourc
esRatio: 0.8
cluster-master    | 26/03/29 15:25:09 INFO BlockManagerMasterEndpoint: Registering block manager cluster-slave-1:39025 with 366.3 MiB RAM, BlockManagerId(2,
 cluster-slave-1, 39025, None)
cluster-master    | 26/03/29 15:25:09 INFO BlockManagerMasterEndpoint: Registering block manager cluster-slave-1:34841 with 366.3 MiB RAM, BlockManagerId(1,
 cluster-slave-1, 34841, None)
cluster-master    | Query: computer science
cluster-master    | Top 10 documents:
cluster-master    | 1. 48227028 A B M Shawkat Ali       25.745640
cluster-master    | 2. 51864384 A Can of Paint  16.355247
cluster-master    | 3. 1443000  A Brief History of Time (film)  14.940743
cluster-master    | 4. 43789955 A Braver Thing  14.716514
cluster-master    | 5. 39271639 A Boy and His Atom      14.502556
cluster-master    | 6. 15812271 A Cabin on the Coast    13.718900
cluster-master    | 7. 58539404 A Birthday      13.606728
cluster-master    | 8. 5259052  A Briefer History of Time (Schulman book)       13.212418
cluster-master    | 9. 651789   A Calculus of Angels    13.129167
cluster-master    | 10. 54088753        A Beautiful Star        12.858641
cluster-master    | Running BM25 search on YARN
cluster-master    | 26/03/29 15:25:22 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where appl
icable
cluster-master    | 26/03/29 15:25:22 INFO SparkContext: Running Spark version 3.5.4
cluster-master    | 26/03/29 15:25:22 INFO SparkContext: OS info Linux, 6.6.87.2-microsoft-standard-WSL2, amd64
cluster-master    | 26/03/29 15:25:22 INFO SparkContext: Java version 1.8.0_442
cluster-master    | 26/03/29 15:25:23 INFO ResourceUtils: ==============================================================
cluster-master    | 26/03/29 15:25:23 INFO ResourceUtils: No custom resources configured for spark.driver.
cluster-master    | 26/03/29 15:25:23 INFO ResourceUtils: ==============================================================
cluster-master    | 26/03/29 15:25:23 INFO SparkContext: Submitted application: bm25-query
cluster-master    | 26/03/29 15:25:23 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name: cores, amount: 1, script
: , vendor: , memory -> name: memory, amount: 1024, script: , vendor: , offHeap -> name: offHeap, amount: 0, script: , vendor: ), task resources: Map(cpus -
> name: cpus, amount: 1.0)
cluster-master    | 26/03/29 15:25:23 INFO ResourceProfile: Limiting resource is cpus at 1 tasks per executor
cluster-master    | 26/03/29 15:25:23 INFO ResourceProfileManager: Added ResourceProfile id: 0
cluster-master    | 26/03/29 15:25:23 INFO SecurityManager: Changing view acls to: root
cluster-master    | 26/03/29 15:25:23 INFO SecurityManager: Changing modify acls to: root
cluster-master    | 26/03/29 15:25:23 INFO SecurityManager: Changing view acls groups to:
cluster-master    | 26/03/29 15:25:23 INFO SecurityManager: Changing modify acls groups to:
cluster-master    | 26/03/29 15:25:23 INFO SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users with view permissions: root; g
roups with view permissions: EMPTY; users with modify permissions: root; groups with modify permissions: EMPTY
cluster-master    | 26/03/29 15:25:23 INFO Utils: Successfully started service 'sparkDriver' on port 45105.
cluster-master    | 26/03/29 15:25:23 INFO SparkEnv: Registering MapOutputTracker
```

Figure 2. Search results for query “computer science”. The engine returns ranked top-10 results with BM25 scores.

```
ilyasgaliev@MSI: ~/bigdata/bi
cluster-master    | 26/03/29 15:26:06 INFO ServerInfo: Adding filter to /stages/pool/json: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 15:26:06 INFO ServerInfo: Adding filter to /storage: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 15:26:06 INFO ServerInfo: Adding filter to /storage/json: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 15:26:06 INFO ServerInfo: Adding filter to /storage/rdd: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 15:26:06 INFO ServerInfo: Adding filter to /storage/rdd/json: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 15:26:06 INFO ServerInfo: Adding filter to /environment: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 15:26:06 INFO ServerInfo: Adding filter to /environment/json: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 15:26:06 INFO ServerInfo: Adding filter to /executors: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 15:26:06 INFO ServerInfo: Adding filter to /executors/json: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 15:26:06 INFO ServerInfo: Adding filter to /executors/threadDump: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 15:26:06 INFO ServerInfo: Adding filter to /executors/threadDump/json: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilt
er
cluster-master    | 26/03/29 15:26:06 INFO ServerInfo: Adding filter to /static: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 15:26:06 INFO ServerInfo: Adding filter to /: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 15:26:06 INFO ServerInfo: Adding filter to /api: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 15:26:06 INFO ServerInfo: Adding filter to /jobs/job/kill: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 15:26:06 INFO ServerInfo: Adding filter to /stages/stage/kill: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 15:26:06 INFO ServerInfo: Adding filter to /metrics/json: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 15:26:06 INFO YarnClientSchedulerBackend: SchedulerBackend is ready for scheduling beginning after waiting maxRegisteredResourc
esWaitingTime: 30000000000(ns)
cluster-master    | Query: artificial intelligence
cluster-master    | Top 10 documents:
cluster-master    | 1. 48227028 A B M Shawkat Ali       9.312347
cluster-master    | 2. 16215956 A Cage of Eagles        8.959491
cluster-master    | 3. 51602191 A Captive in the Land (novel)   8.088543
cluster-master    | 4. 52968927 A Bag of Marbles (2017 film)    7.749544
cluster-master    | 5. 71529300 A Boyfriend for My Wife (2022 film)     7.580110
cluster-master    | 6. 19959644 A Book on Angling       7.555742
cluster-master    | 7. 21950055 A Bunch of Amateurs     7.287981
cluster-master    | 8. 69089425 A Brightness Long Ago   6.517343
cluster-master    | 9. 15131446 A Change of Seasons (film)      6.220150
cluster-master    | 10. 21004590        A Brief History of the Future   6.097087
cluster-master exited with code 0
cassandra-server  | INFO  [ScheduledTasks:1] 2026-03-29 16:20:01,199 ColumnFamilyStore.java:1052 - Enqueuing flush of system.compaction_history, Reason: MEM
TABLE_PERIOD_EXPIRED, Usage: 11.498KiB (0%) on-heap, 0B (0%) off-heap
cassandra-server  | INFO  [ScheduledTasks:1] 2026-03-29 16:20:01,207 ColumnFamilyStore.java:1052 - Enqueuing flush of system.sstable_activity, Reason: MEMTA
BLE_PERIOD_EXPIRED, Usage: 11.655KiB (0%) on-heap, 0B (0%) off-heap
cassandra-server  | INFO  [PerDiskMemtableFlushWriter_0:5] 2026-03-29 16:20:01,253 Flushing.java:153 - Writing Memtable-sstable_activity@757110047(280B seri
alized bytes, 35 ops, 11.655KiB (0%) on-heap, 0B (0%) off-heap), flushed range = [min(-9223372036854775808), max(9223372036854775807))
cassandra-server  | INFO  [PerDiskMemtableFlushWriter_0:6] 2026-03-29 16:20:01,256 Flushing.java:153 - Writing Memtable-compaction_history@990226865(2.357Ki
B serialized bytes, 8 ops, 11.498KiB (0%) on-heap, 0B (0%) off-heap), flushed range = [min(-9223372036854775808), max(9223372036854775807))
cassandra-server  | INFO  [PerDiskMemtableFlushWriter_0:5] 2026-03-29 16:20:01,257 Flushing.java:179 - Completed flushing /var/lib/cassandra/data/system/sst
able_activity-5a1ff267ace03f128563cfae6103c65e/nb-1-big-Data.db (1.606KiB) for commitlog position CommitLogPosition(segmentId=1774795631932, position=329498
70)
```

Figure 3. Search results for query “artificial intelligence”. The top-10 list is produced successfully on YARN.

```
ilyasgaliev@MSI: ~/bigdata/bi
cluster-master    | 26/03/29 17:38:40 INFO ServerInfo: Adding filter to /stages: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 17:38:40 INFO ServerInfo: Adding filter to /stages/json: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 17:38:40 INFO ServerInfo: Adding filter to /stages/stage: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 17:38:40 INFO ServerInfo: Adding filter to /stages/stage/json: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 17:38:40 INFO ServerInfo: Adding filter to /stages/pool: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 17:38:40 INFO ServerInfo: Adding filter to /stages/pool/json: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 17:38:40 INFO ServerInfo: Adding filter to /storage: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 17:38:40 INFO ServerInfo: Adding filter to /storage/json: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 17:38:40 INFO ServerInfo: Adding filter to /storage/rdd: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 17:38:40 INFO ServerInfo: Adding filter to /storage/rdd/json: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 17:38:40 INFO ServerInfo: Adding filter to /environment: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 17:38:40 INFO ServerInfo: Adding filter to /environment/json: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 17:38:40 INFO ServerInfo: Adding filter to /executors: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 17:38:40 INFO ServerInfo: Adding filter to /executors/json: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 17:38:40 INFO ServerInfo: Adding filter to /executors/threadDump: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 17:38:40 INFO ServerInfo: Adding filter to /executors/threadDump/json: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilt
er
cluster-master    | 26/03/29 17:38:40 INFO ServerInfo: Adding filter to /static: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 17:38:41 INFO ServerInfo: Adding filter to /: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 17:38:41 INFO ServerInfo: Adding filter to /api: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 17:38:41 INFO ServerInfo: Adding filter to /jobs/job/kill: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 17:38:41 INFO ServerInfo: Adding filter to /stages/stage/kill: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 17:38:41 INFO ServerInfo: Adding filter to /metrics/json: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
cluster-master    | 26/03/29 17:38:41 INFO YarnSchedulerBackend$YarnSchedulerEndpoint: ApplicationMaster registered as NettyRpcEndpointRef(spark-client://Ya
rnAM)
cluster-master    | 26/03/29 17:38:43 INFO YarnClientSchedulerBackend: SchedulerBackend is ready for scheduling beginning after waiting maxRegisteredResourc
esWaitingTime: 30000000000(ns)
cluster-master    | Query: galievtesttoken
cluster-master    | Top 10 documents:
cluster-master    | 1. 99999999 GalievTestToken 14.329413
cluster-master    | 26/03/29 17:38:49 WARN Dispatcher: Message RequestMessage(172.18.0.3:33926, NettyRpcEndpointRef(spark://CoarseGrainedScheduler@cluster-m
aster:37875), RemoveExecutor(2,Unable to create executor due to Exception thrown in awaitResult: )) dropped due to sparkEnv is stopped. Could not find Coars
eGrainedScheduler.
cluster-master exited with code 0
cassandra-server  | INFO  [IndexSummaryManager:1] 2026-03-29 17:43:46,690 IndexSummaryRedistribution.java:103 - Redistributing index summaries
cassandra-server  | INFO  [IndexSummaryManager:1] 2026-03-29 17:43:46,692 IndexSummaryRedistribution.java:115 - Beginning redistribution of index summaries
for 28 sstables with memory pool size 196 MiB; current spaced used is 6.875991821289062E-4 MiB
cassandra-server  | INFO  [IndexSummaryManager:1] 2026-03-29 17:43:46,699 IndexSummaryRedistribution.java:268 - index summaries: downsample: 0, force resamp
le: 0, upsample: 0, force upsample: 0
cassandra-server  | INFO  [IndexSummaryManager:1] 2026-03-29 17:43:46,704 IndexSummaryRedistribution.java:154 - Completed resizing of index summaries; curre
nt approximate memory used: 6.875991821289062E-4 MiB, time spent: 13ms

v View in Docker Desktop   o View Config   w Enable Watch
```

Figure 4. The optional add_to_index flow completes successfully, and the newly inserted document becomes the top result for query “galievtesttoken”.

2.3 Interpretation of the Retrieved Results

For the query “computer science”, the engine returned a ranked list headed by “A B M Shawkat Ali”, followed by “A Can of Paint” and “A Brief History of Time (film)”. For the query “artificial intelligence”, the top results were again produced correctly by the BM25 ranker. While some high-ranked documents are not semantically strong matches in a human sense, this is acceptable for a baseline BM25 system on a noisy general-purpose corpus and reflects the fact that lexical matching can surface documents with overlapping high-weight terms without semantic understanding.

The most important validation for the optional extension is more direct: after add_to_index.sh inserted 99999999_GalievTestToken.txt and rebuilt the index, the query “galievtesttoken” returned that document as the top hit. This confirms that the index refresh path and Cassandra reload path are working.

3. Reflections and Final Notes

- The final implementation satisfies the assignment architecture: PySpark for preparation, Hadoop MapReduce for indexing, Cassandra for storage, and Spark/YARN for retrieval.
- A key debugging improvement was adding visibility to the long /data upload phase through progress logging.
- A practical engineering trade-off is that the optional add_to_index path rebuilds the index rather than performing a fine-grained incremental update. This is slower, but it is simple, deterministic, and correct for the assignment scope.